

Airwallex: Leveraging deep collaboration with Google Cloud to provide a global payments platform that powers cross- border business

With Google Cloud
Airwallex is achie

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

Founded in Melbourne and with



About Airwallex

Founded in Melbourne and with engineering hubs in Amsterdam, Melbourne, San Francisco, Shanghai, Singapore, Hong Kong, and Sydney, Airwallex is a global payments platform with a mission to empower businesses of all sizes to grow without borders, and by doing so, contribute to the global economy. With technology at its core, Airwallex has built a financial infrastructure and platform to help businesses manage online payments, treasury, and payout globally, without the constraints of the traditional financial system. Airwallex has secured over US\$800 million since it was established in 2015, and is backed by world-

geographic cover
reliability, latency
data consistency
reduction, and sp
innovation that s
vision of being a
economic infrast
that enables busi
operate across b
anywhere at any

leading investors. Today, the business operates
with a team of almost 1,000 employees across
20 locations globally. For more information,
please visit www.airwallex.com
(<https://www.airwallex.com>).

Industries: Financial Services &
Insurance

Location: Australia

Google Cloud results

- Meets
business
requirements
by recording
availability
over 99.95%
- Ensures
latency is
well below
200
milliseconds,
above which
throughput
of the

Employ
reliab
engin
impro
application
reliability

(<https://cloud.google.com/>), [Kubernetes Engine](https://cloud.google.com/kubernetes-engine/)

Google Cloud (<https://cloud.google.com/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://cloud.google.com/products/operations>)

Anthos (<https://cloud.google.com/anthos/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Operations

(<https://cloud.google.com/products/operations>)

Contact us (/contact)

distributed
system that
requires
strong
consistency
will drop
significantly

- Enables the business to use a data warehouse and data lake to enable analysis of data-governing team performance and the impact of product changes on conversion rates—driving better performance and customer experience
- Scales to support an increase of about 30x in transactions

per second
(tps) to
50,000 tps

Founded in 2015 in Melbourne, Australia, and with offices across Asia, Europe, and the United States today, [Airwallex](https://www.airwallex.com/au) (<https://www.airwallex.com/au>) was born when two friends, Jack Zhang and Max Li, started a cafe and immediately experienced how high foreign exchange fees and banking costs could erode profit margins. Together with University of Melbourne friends Lucy Liu, Xijing Dai, and Ki-Lok Wong, Jack and Max launched a platform that aimed to empower businesses of all sizes to grow and contribute to the global economy.

By late 2021, Airwallex had grown to almost 1,000 employees and 20 locations worldwide, offering products that enabled businesses to open domestic and international accounts, create multi-currency payment cards, synchronize transactions with Xero online accounting software, undertake low-cost and high speed international payments, and create simple links for instant payment. The business also operates a number of API-powered services that allow customers to accept payments online, build the best foreign exchange rates into customer pricing and transactions, manage and issue cards, create accounts and store and manage funds worldwide, program domestic and international payments, and build their own payment and financial products.

To disrupt and transform the traditional payments and banking sector, Airwallex is embracing new technologies with a particular focus on modernized platform architectures that power fast, reliable, and scalable applications designed to help businesses overcome their financial challenges. Three or four years ago, the organization elected to move away from its incumbent cloud service that could not easily support its plans to continue expanding into new markets or support compliance with regulations or payment card-handling information security standards such as PCI DSS (Payment Card Industry Data Security Standard).

Given how essential PCI DSS is for the payment acceptance and card issuance business lines, Airwallex needed to ensure that its infrastructure and other PCI DSS-related applications met stringent industry assessments for compliance. With Google Cloud (<https://cloud.google.com/>), the company was able to leverage an established security infrastructure that covered all encryption usage concerns. It was also really easy for the company to provide documentation during audits to support claims of Airwallex's systems being built on top of secure and trustworthy technologies.

Outstanding functionality and world class security

“We started to explore Google Cloud and undertook a proof of concept exercise,” explains Ryan Cao, VP Development/Principal Architect at Airwallex. “We found the documentation clear and easy to follow, and when we began testing individual products and

services, we found they had outstanding functionality combined with world-class security—a commitment that also extended to multilayered security at its own data centers. This process taught us a lot about how we should secure our own and our customers' data.”

This education was particularly valuable for a business that relies heavily on security to build and retain trust with customers that use Airwallex to manage millions of dollars in payments, make domestic and international transfers, and sell globally.

Airwallex needed to complement high security and compliance with domestic and international regulations with reliability, low latency, and manageable costs. Furthermore, as a startup, the business had to focus on the application layer of its offerings rather than on dedicating time and resources to infrastructure.

“We have to go
up from 1,500
transactions per
second to
50,000
transactions per
second without

compromising
performance,
availability, or
consistency, and
Google
Kubernetes
Engine enables
us to do this
with ease.”

—Ryan Cao, VP
Development/Principal
Architect, Airwallex

Latency at least 50% lower

Google Cloud’s network and global portfolio of data centers was also particularly compelling for Airwallex, and the organization’s testing revealed latency across the Google network was at least 50% lower than the latency across the networks used by other cloud providers. “As a business, we really need that low latency, because that is how we ensure database throughput while its distributed transactions stay consistent,” explains Ryan. “Low latency means we can receive and offer low rates to customers without applying a high spread to account for the rate fluctuations that occur due to the fast-moving nature of financial markets. It also ensures we can provide data consistency without

being concerned about issues that occur if latency does exceed about 200 milliseconds.”

Airwallex has now expanded its use of Google Cloud to more than 50% of the provider’s products and services, and rates [Google Kubernetes Engine](#) (/kubernetes-engine) as almost equal to the Google Cloud network in importance to the business. The organization uses containerization to run its microservices architecture and its applications reliably and flexibly, with only a few Kafka and Postgres clusters and virtual machines running outside the Google Kubernetes Engine environment.

“It was also relatively easy for us to migrate our workload to Google Cloud as we were using Kubernetes to run our containerized workload. We used Megaport to bridge our applications in Google Cloud to the other cloud vendor, so the migrated services can access other system components there. Payment acceptance and card issuance products were built entirely on Google Cloud with all the Google Cloud unique features being taken into consideration in the system design from day one,” adds Ryan.

Seamless scalability

Google Kubernetes Engine also enables Airwallex to seamlessly support demand spikes that occur on its platform when customers run commercial events, with the organization able to spin up new nodes in less than one minute. “We have to go up from 1,500 transactions per second to 50,000 transactions per second without compromising performance,

availability, or consistency, and Google Kubernetes Engine enables us to do this with ease,” says Ryan.

The managed Kubernetes product and Google Cloud more generally also enables Airwallex to meet the short development and release cycle times required to constantly change its applications. The cloud platform also ensures the business enables engineers to own and experiment with changes before releasing them to production. “Being able to accelerate and deliver change in this way is why we adopted microservices and containerization,” explains Ryan. “Google Kubernetes Engine, combined with the [Anthos](#) (/anthos)-managed application and infrastructure deployment platform and the Google Cloud network, enables us to release changes simultaneously to multiple clusters in multiple data centers around the world.”

Easy data warehouse creation

Airwallex has turned to [BigQuery](#) (/bigquery) to create and run a data warehouse and data lake that enables the analysis of large volumes of data without requiring the business to invest in data infrastructure and internal expertise in star schema modeling, fact and dimension table design and other warehouse creation techniques.

“BigQuery enables us to drive decisions based on data collected through our Salesforce customer relationship management system and through Google Analytics,” says Ryan. “For example, we use the product to analyze how changes to our products impact conversion rates, to quickly and efficiently

reconcile transactions and files we receive from banks and our partners, and to manage the performance of our commercial team by helping them forecast customer transaction volumes and compare them to our business's goals.

"This enables us to quickly identify opportunities to improve execution and optimize the experience we provide to our customers."

Using BigQuery as a data lake also gives Airwallex a repository for all application data and change history, with this data now easily available for the business to train machine learning models.

Airwallex is now ramping up the use of machine learning and AI to combat attempted fraud and account takeover and to detect suspicious transactions and behaviors, while the business already uses predictive technologies and models to ensure each of its payment network nodes holds sufficient funds at any time to meet customer requirements.

Maintenance downtime cut from five minutes to 10 seconds per quarter

Cloud SQL (/sql), Google Cloud's fully managed relational database service for MySQL, PostgreSQL, and SQL Server, houses all transactional data and operates as a cornerstone of the Airwallex infrastructure, delivering the high availability needed to support a 24-hour a day business. For example, Airwallex uses Cloud SQL for PostgreSQL (/sql/docs/postgres), and with so many database

instances, the standard ones use 4-16 vCPU, 15-64 GB memory, and 10-300 GB SSD storage. Database downtime means that transactions could not be processed during maintenance, so the business worked closely with Google Cloud to shorten quarterly maintenance downtime to 10 seconds or less.

The business now records latency ranging from 70 milliseconds to 130 milliseconds across the Google Cloud global network, with reliability that exceeds business requirements and a transparent network maintenance schedule that minimizes disruption. This complements availability across the Airwallex technology stack of 99.95%, a number the organization aims to work with Google Cloud and its vendors to increase to 99.999%.

“We need to
have a platform
that supports
our international
ambitions and
help us go for
growth—we are
in a highly
competitive
market in the

fintech world,
and our services
need to be
outstanding
across
geographic
coverage,
reliability,
latency, security,
data
consistency,
cost, and speed
to innovate.
Google Cloud
helps us deliver
on all of these.”

—Ryan Cao, VP
Development/Principal
Architect, Airwallex

Implementing SRE with Google Cloud

Airwallex is now working with [Google Cloud Professional Services](#) (/consulting) to establish a site reliability engineering (SRE) team to improve the reliability of its services across cloud infrastructure, application architecture, and application operations. “We can learn a lot from Google Cloud because it

pioneered site reliability engineering and has accumulated considerable knowledge about how to form a successful SRE team,” says Ryan. “We have a very close relationship with the Professional Services team and they are proactive about diving into our solution architecture and providing solutions to any issues.”

The cloud provider’s SRE offerings enable Airwallex to balance speed and reliability, with the Professional Services team helping the business deploy the Google Cloud operations suite (/products/operations) to monitor its applications and receive alerts when potential issues arise.

“Overall, Google Cloud is helping us execute our vision as a business, which is to become a global financial cloud that enables businesses to operate anywhere at any time,” concludes Ryan. “We need to have a platform that supports our international ambitions and help us go for growth—we are in a highly competitive market in the fintech world and our services need to be outstanding across geographic coverage, reliability, latency, security, data consistency, cost, and speed to innovate. Google Cloud helps us deliver on all of these.”